Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | sahaj-bhadja | sahaj-bhadja | sahaj-bhadja

Summary

Software engineer with 2+ years building scalable distributed services, AI-assisted workflows, data pipelines, and full-stack applications using Python, C++, Java, TypeScript, SQL, cloud infrastructure, automated testing, and Linux.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant **PostgreSQL, Azure SQL, and Blob Storage** services for 500+ users, optimizing schemas and indexed queries to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant **FastAPI orchestration** for 1,000+ monthly AI jobs using explicit states, retries, evaluation checkpoints, and failure isolation, tripling throughput while strengthening production reliability and debugging visibility.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Python APIs and React workflows** integrating GPT-4o, Claude, Qwen-VL, and OCR to evaluate 10,000+ daily submissions while maintaining stable performance and 90%+ grading consistency.
- Designed concurrent AWS RDS-backed processing for 1,000+ files per batch using multithreading, secure ingestion, and fault-tolerant execution, reducing model evaluation turnaround by 60% under peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based Decision Transformer environment for autonomous-driving simulations, processing 100,000+ trajectory logs and improving long-horizon planning by 18% through reproducible **machine-learning pipelines and systematic evaluation**.
- Architected a spatial-temporal encoder combining HD-map geometry and historical agent trajectories through multi-head attention, improving closed-loop simulation stability by 15% across diverse, complex real-world scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform **Python microservices** and 20+ automated PyTest and Mockito suites through Jenkins CI/CD, improving release reliability across Windows, Linux, and macOS production environments.
- Optimized REST API and ETL integrations using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-pipeline efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a MERN platform with REST APIs, WebSockets, and eight modular Node.js services, supporting 2M+ synchronized records while translating complex requirements into maintainable real-time application workflows.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, documentation, and defect resolution across frequent production releases.

Skills

Languages & Systems: Python, C++, Java, JavaScript, TypeScript, SQL, C, Bash, Linux, Unix, data structures, algorithms, object-oriented programming, concurrency, multithreading, distributed systems, system design, networking; **Applications & Data:** React, Node.js, FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, information retrieval, ETL, data pipelines, NLP, machine learning, deep learning; **Cloud & Engineering:** GCP, AWS, Azure, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, automated testing, debugging, performance optimization, monitoring, Agile/Scrum, AI-assisted development, prompt engineering, code review

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, Algorithms*
- Implemented six **CPU scheduling algorithms** with dynamic metrics and Gantt visualization, validating preemptive and non-preemptive execution across 1,000+ process combinations while reducing algorithm-debugging time by 20%.

Big-Brain Live | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built an AI data platform for 200+ users with secure ingestion, vector indexing, **semantic information retrieval**, and collaborative workflows, maintaining 95% query accuracy across 50,000+ documents.

CaptionCraft Live | *Python, PyTorch, Transformers, Flask, Gradio*
- Benchmarked BLIP, GIT, ViT-GPT-2, and custom transformer models on 10,000 images and 100,000 captions, improving **model evaluation accuracy** from 81% to 96% through systematic experimentation.